Basic Principles of Medicine 2
ER Lecture 18

Female Reproductive Physiology

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Recommended reading

Medical Physiology - Principles for Clinical Medicine. Rhoades & Bell, 6e.

Chapter 37 – Female Reproduction; pages 791 - 809
https://meded-lwwhealthlibrary-com.sgu.idm.oclc.org/content.aspx?sectionid=253092173&bookid=3211

Chapter 38 – Fertilization, and Pregnancy; pages 810-828
https://meded-lwwhealthlibrary-com.sgu.idm.oclc.org/content.aspx?sectionid=253092355&bookid=3211

Objectives

SOM.MK.II.BPM2.1.ER.1.PHYS.1074.Describe the role of human female in reproduction

SOM.MK.II.BPM2.1.ER.1.PHYS.1075 Explain the roles of GnRH, FSH, LH, estradiol, and inhibin in oogenesis and follicular maturation in females at puberty, the start of ovarian cycle

SOM.MK.II.BPM2.1.ER.1.PHYS.1076. Describe the hormonal regulation of estrogen and progesterone biosynthesis and secretion by the ovary.

Objectives

- SOM.MK.II.BPM2.1.ER.1.PHYS.1077.Describe the hormonal changes, follicular development, ovulation formation & atresia of corpus luteum throughout the ovarian cycle
- SOM.MK.II.BPM2.1.ER.1.PHYS.1078. List the major target organs for estrogen action and describe its effects on each.
- SOM.MK.II.BPM2.1.ER.1.PHYS.1079. List the principal physiological actions of progesterone, its major effects on target organs.

Objectives

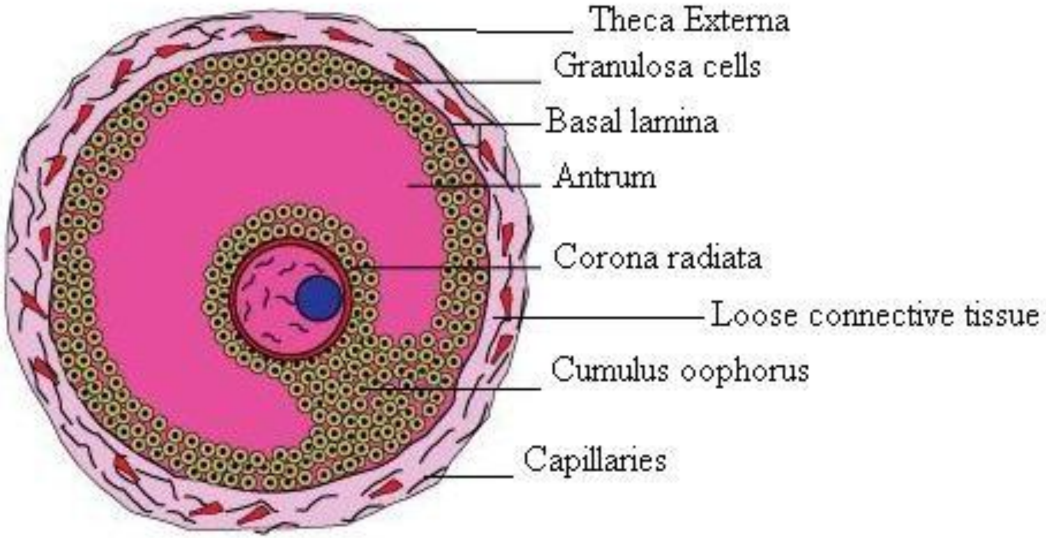
- SOM.MK.II.BPM2.1.ER.1.PHYS.1080. Describe how the changes in ovarian steroids produce the proliferative and secretory phases of the uterine endometrium and menstruation and the changes in basal body temperature during the menstrual cycle

- SOM.MK.II.BPM2.1.ER.1.PHYS.1081. Explain the age-related changes in the hypothalamo-pituitary-goandal axis in females that lead to puberty, reproductive maturity, and reproductive senescence (menopause).

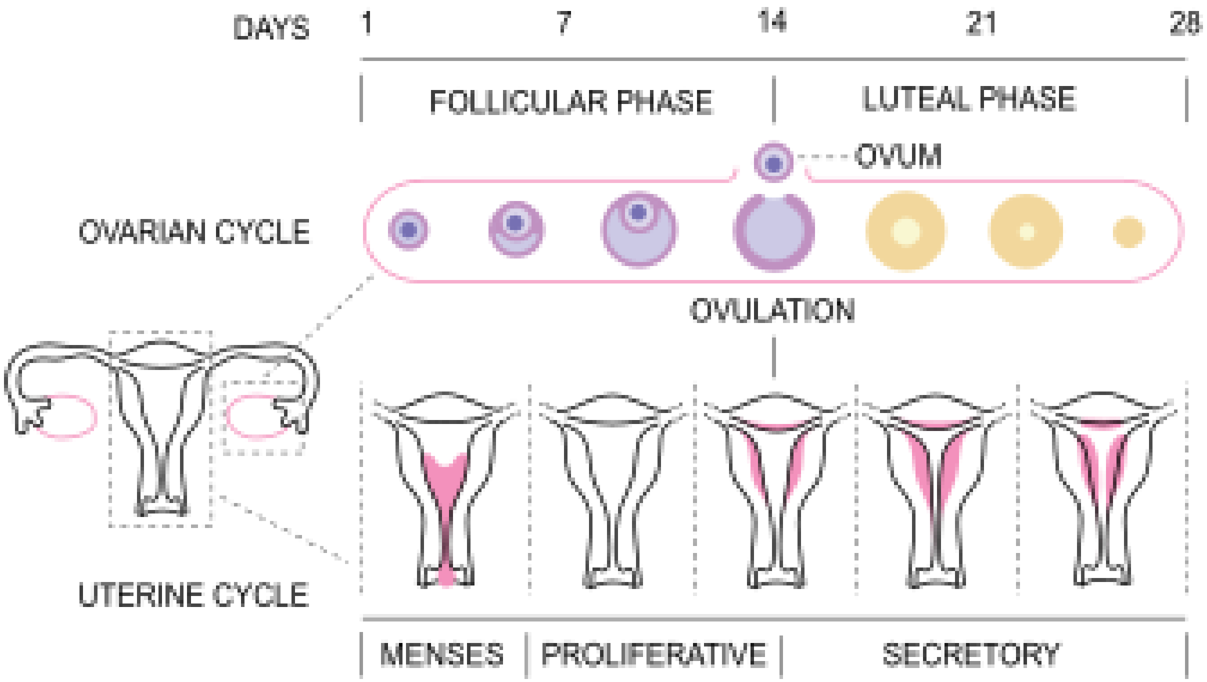
Objectives

- SOM.MK.II.BPM2.1.ER.1.PHYS.1082. Explain the physiological basis of steroid hormone contraception.
- SOM.MK.II.BPM2.1.ER.1.PHYS.1083. Explain age related changes in hypogonadism in females.

Review: Granulosa cells



Review: Menstrual cycle



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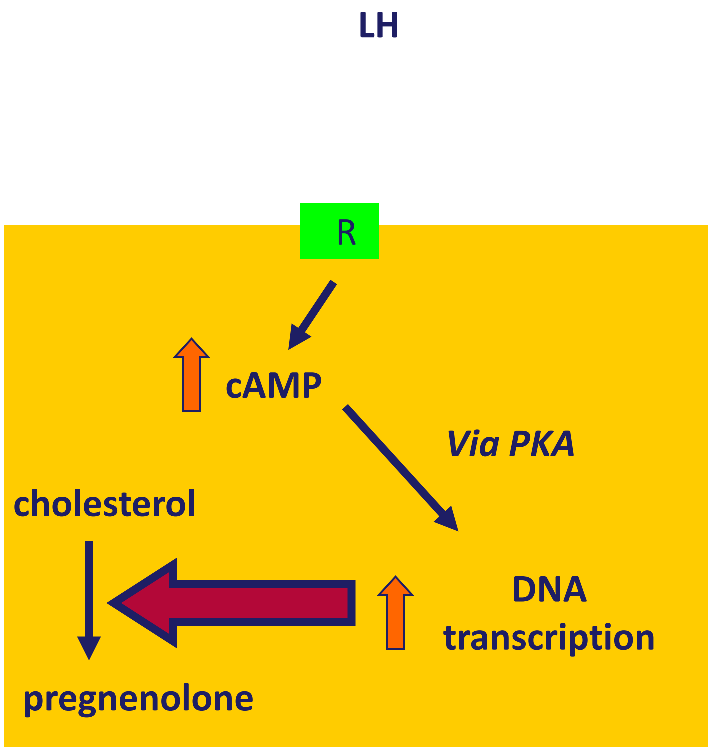
Granulosa & Theca cells

Thecal Cell - Ovary



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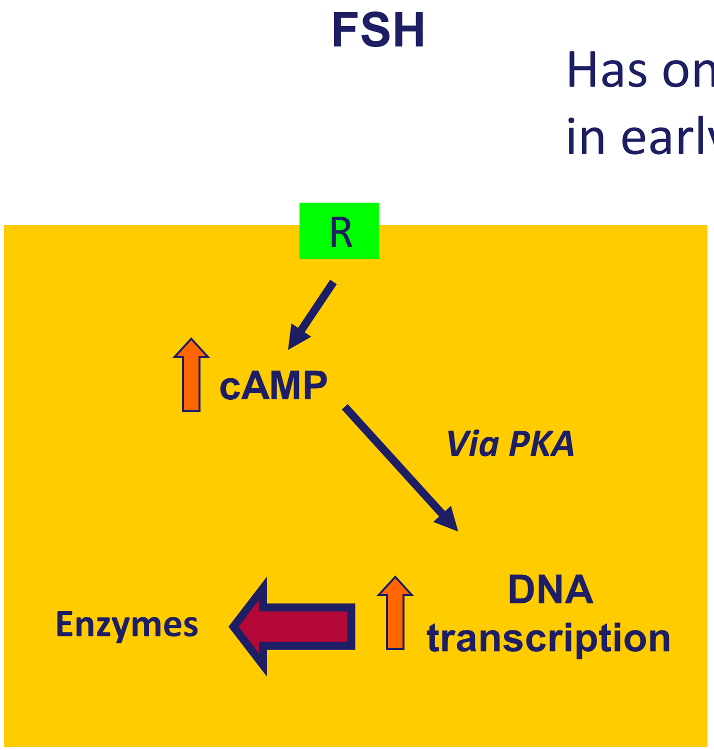
Granulosa Cell – early to mid follicular phase

Has only FSH receptor in early phase



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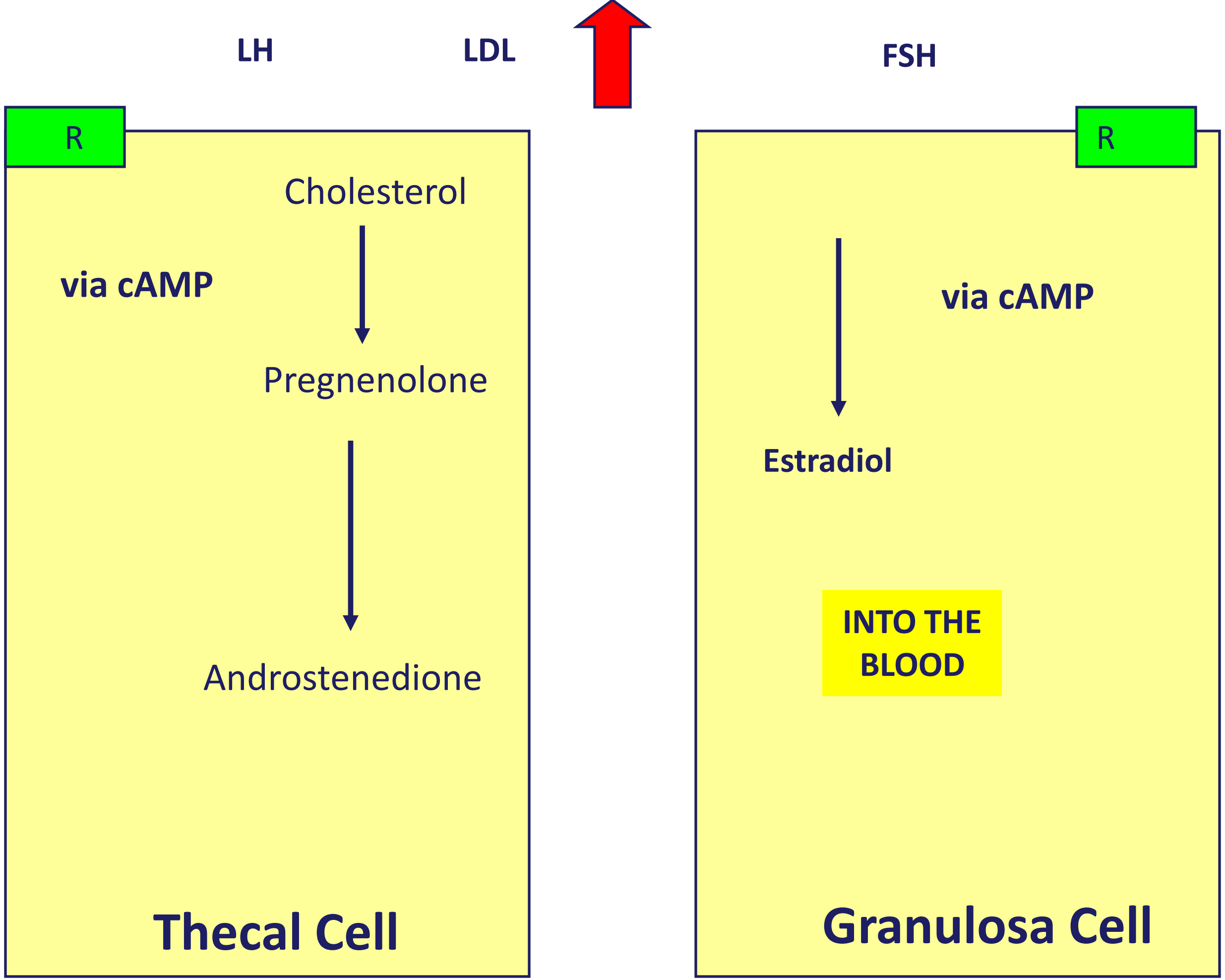

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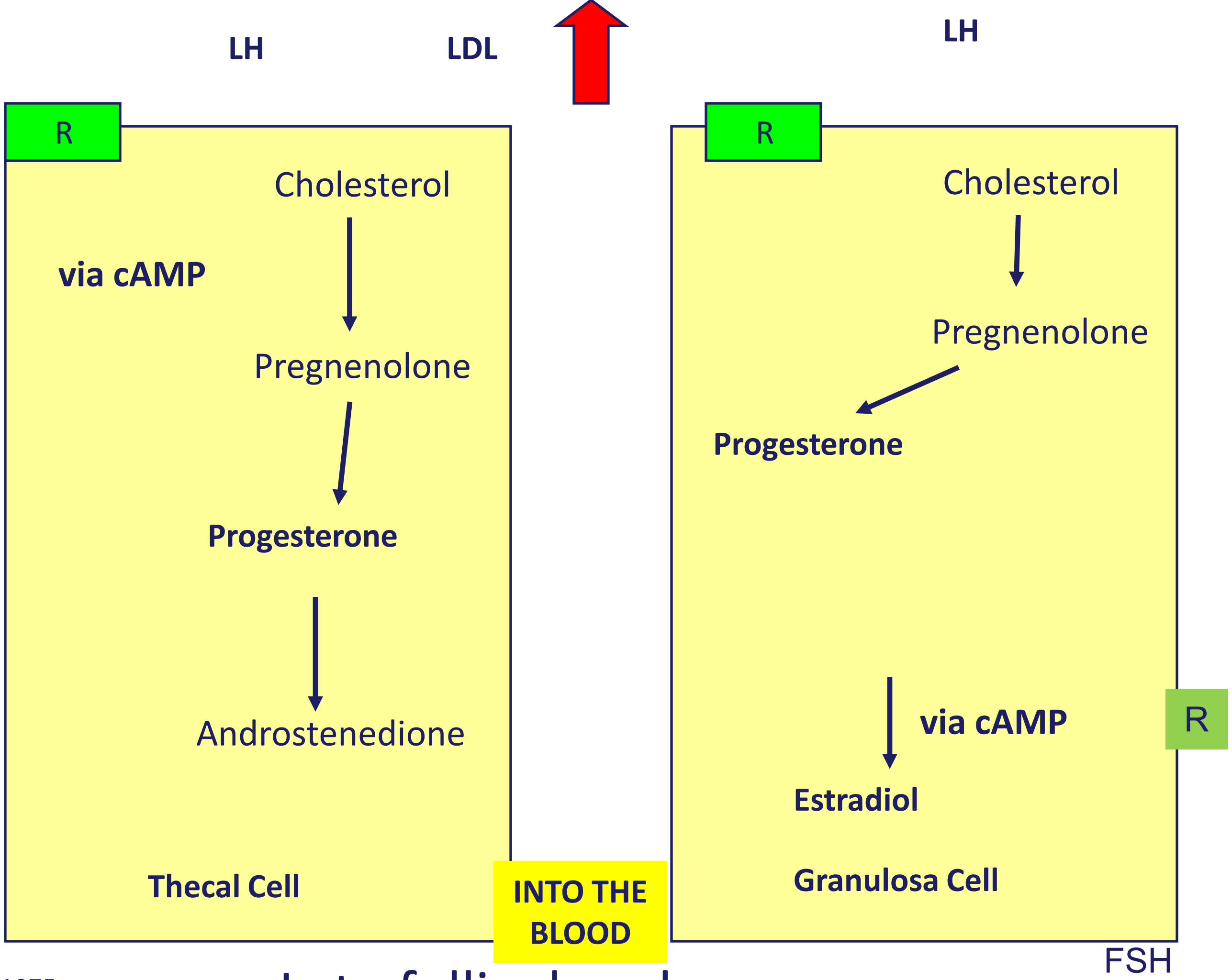
Two cell theory- early to mid follicular stage



Late follicular phase

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Summary two cell theory

Early follicular phase

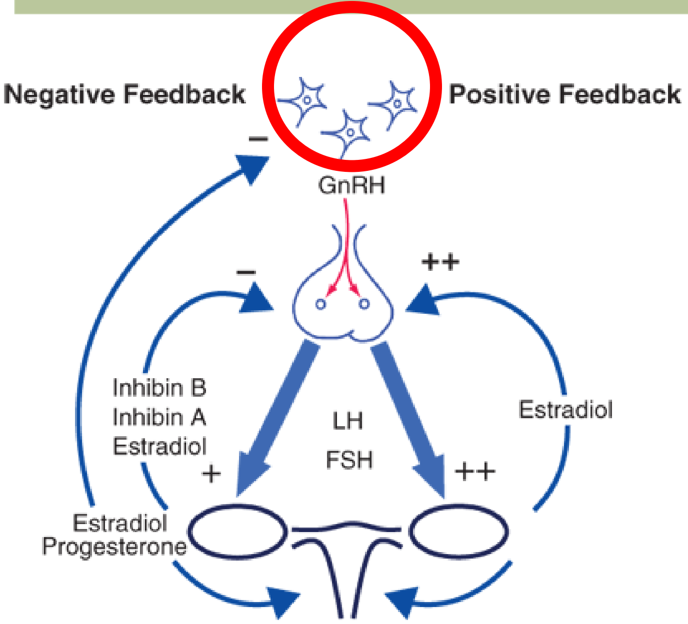
- LH binds to its receptor on the thecal cell causing an increase in uptake of LDL’s into the cell as well as stimulating the conversion of cholesterol into pregnenolone.
- This leads to the production of androstenedione.
- This androstenedione leaves the thecal cell and moves into the granulosa cell.
- The binding of FSH to its receptor on the granulosa cell increases the production of the aromatase enzyme that converts androstendione into estradiol. This estradiol then diffuses into the blood

Late follicular phase:

- Granulosa cell acquires an LH receptor under the influence of FSH
- Granulosa cell can now synthesize progesterone from cholesterol

Female Sex Hormone Physiology

Feedback mechanisms- hypothalamus



Source: J.L. Jameson, A.S. Fauci, D.L. Kasper, S.L. Hauser, D.L. Longo, J. Loscalzo: Harrison's Principles of Internal Medicine, 20th Edition

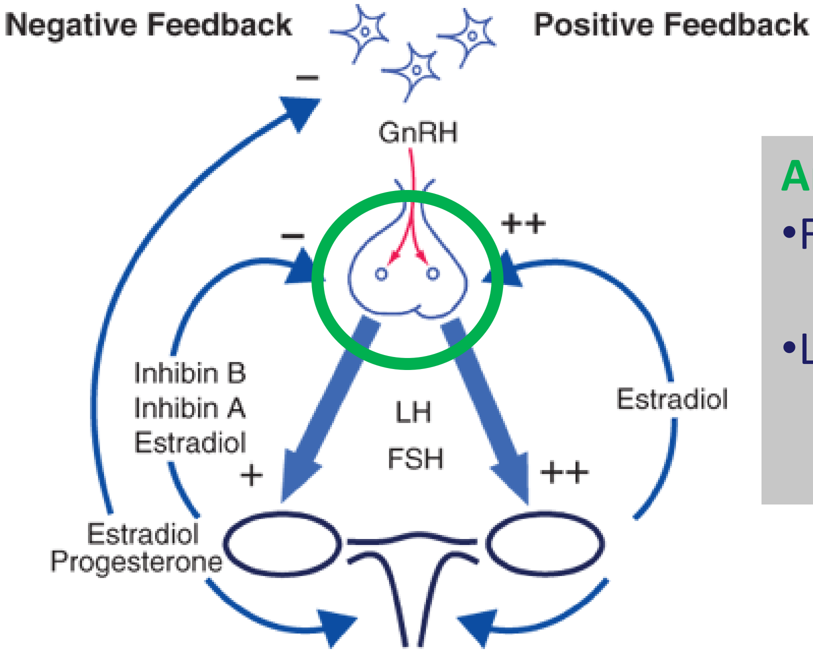
Hypothalamus

- GnRH pulse generator
- Changes in GnRH secretion (frequency & amplitude) to differentially modulate LH and FSH.

- Slow freq GnRH → FSH
- ↑ freq & amp GnRH → LH

Harrison's Principles of Internal Medicine, 20e > Disorders of the Female Reproductive System
https://accessmedicine.mhmedical.com/ViewLarge.aspx?figid=192287780&gbosContainerID=0&gbosid=0&groupID=0 , www.ncbi.nlm.nih.gov/books/NBK279070, Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 13e

Feedback mechanisms- anterior pituitary



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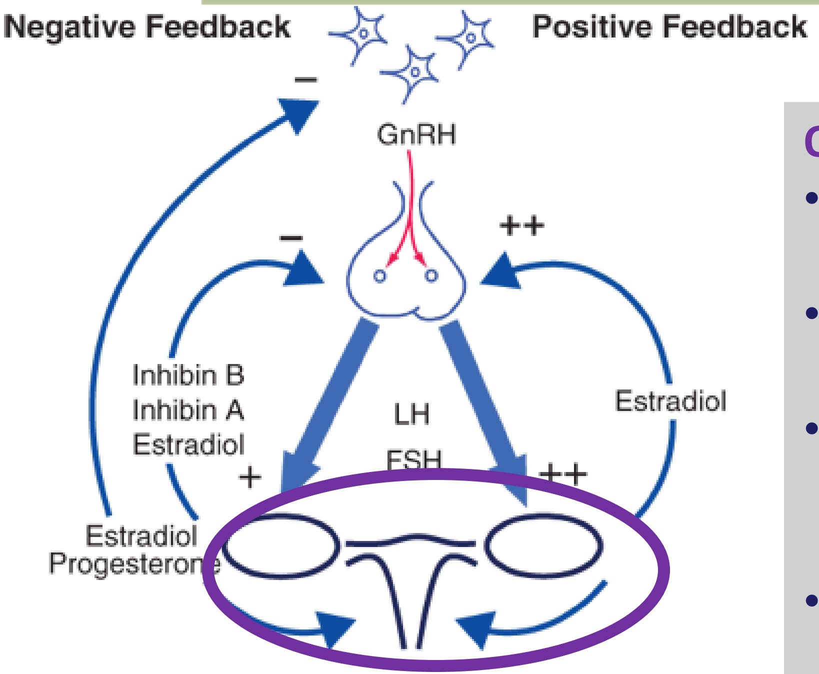
Anterior Pituitary

- FSH → effect granulosa cells
- LH → thecal cells

Feedback mechanisms- ovary



Source: J.L. Jameson, A.S. Fauci, D.L. Kasper, S.L. Hauser, D.L. Longo, J. Loscalzo: Harrison's Principles of Internal Medicine, 20th Edition

Ovary

- Granulosa cells → estrogen
- Thecal cells → testosterone
- Corpus luteum → estrogen & progesterone
- Ovary- inhibin

Harrison's Principles of Internal Medicine, 20e > Disorders of the Female Reproductive System
https://accessmedicine.mhmedical.com/ViewLarge.aspx?figid=192287780&gbosContainerID=0&gbosid=0&groupID=0

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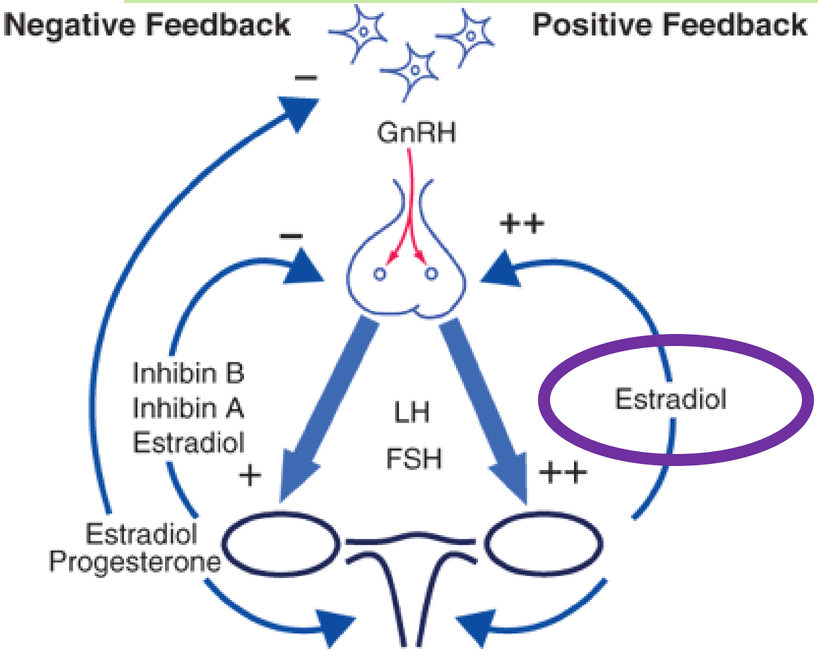
Feedback mechanisms- estrogen



Source: J.L. Jameson, A.S. Fauci, D.L. Kasper, S.L. Hauser, D.L. Longo, J. Loscalzo: Harrison's Principles of Internal Medicine, 20th Edition

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NOTE!!

Estrogen normally has *negative feedback.*

However just prior to ovulation: estrogen switches to <u>positive feedback</u> and mediates LH surge (in later slides)

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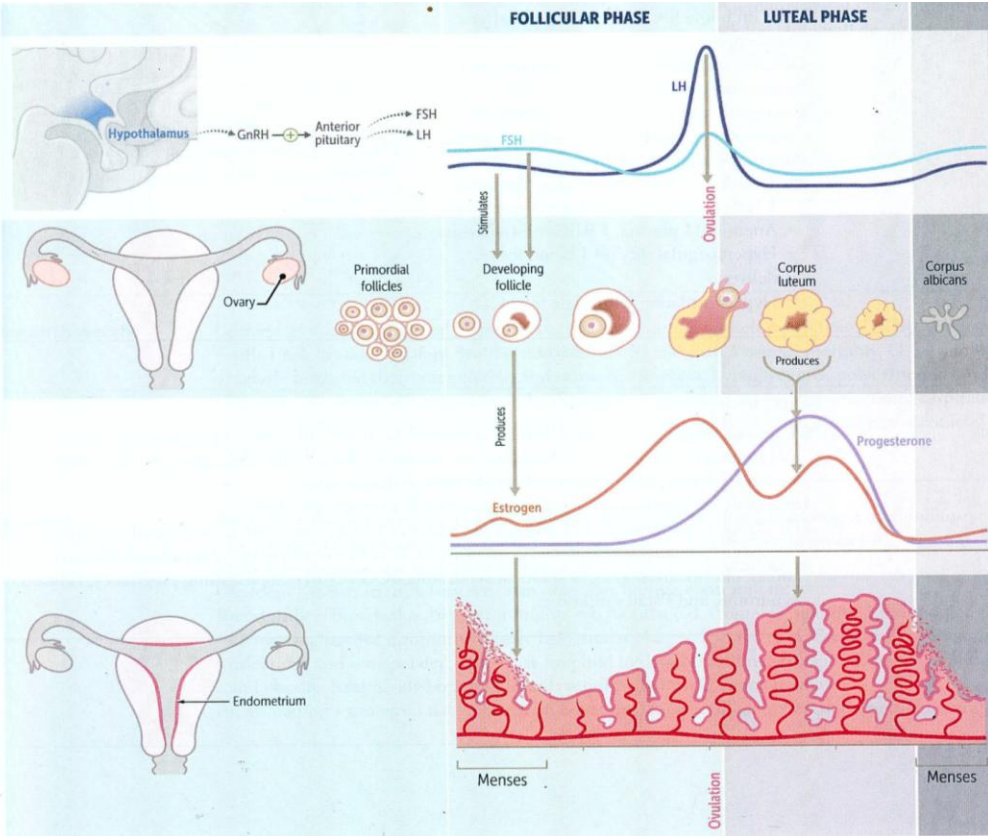
3 organs involved- brain, ovary & genitalia



First Aid 2016 USLME Step 1. McGraw Hill

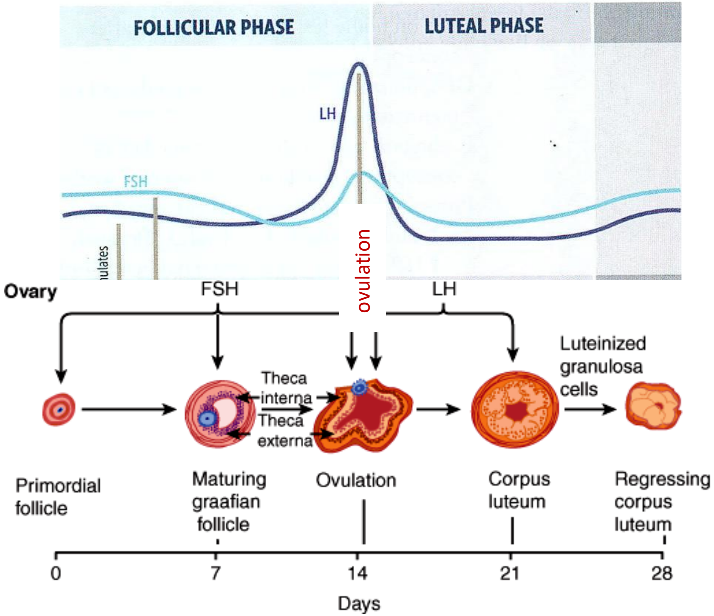
Hypothalamus: GnRH
Pituitary: LH and FSH

Ovary:
Egg- granulosa & theca cells
Corpus luteum

Uterus & Vagina:
Changes in preparation for fertilization & pregnancy

Ovarian Cycle: Follicular & Luteal
(Phases separated by ovulation)

SOM.MK.II.BPM2.1.ER.1.PHYS.1075



Follicular phase
- Development of follicle → FSH
- Follicular phase from menses until ovulation ~ 14 days. (interval variable)

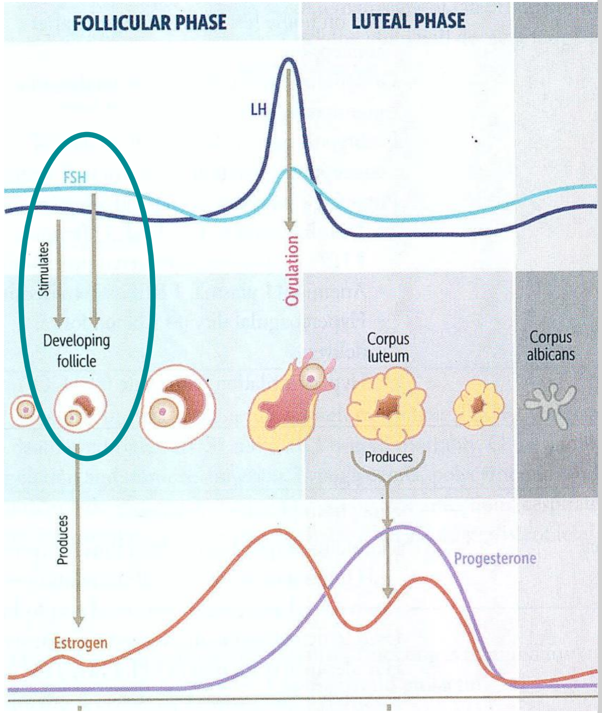
Luteal phase
- Development of corpus luteum
- Luteal phase =14 days (fixed by finite lifespan of corpus luteum)

Length of cycle is 28- 35 days

In general, FSH stimulates follicular growth in the ovary
LH, drives ovulation & progesterone secretion from the developing corpus luteum

Laboratory Medicine: The Diagnosis of Disease in the Clinical Laboratory Chapter 22: Endocrine system
Image modified from Pathophysiology of Disease: An Introduction to Clinical Medicine, 7e Disorders of the Female Reproductive Tract, and First Aid 2016 USLME Step 1. McGraw Hill.

Menstrual cycle: Follicle stimulating hormone (FSH)



Modified from First Aid 2016 USLME Step 1. McGraw Hill

FSH increases in early follicular phase (&late luteal)

- Important for follicular development
- Due to:
  - Decreased inhibin & estrogen negative feedback on pituitary
  - Decreased progesterone & estrogen negative feedback on hypothalamus

FSH declines until ovulation

- Due to increased estrogen secretion by granulosa cells of growing follicle.
- FSH decrease important for selection of dominant follicle.

Decreased FSH in luteal phase (not needed for follicle development in this phase)

Laboratory Medicine: The Diagnosis of Disease in the Clinical Laboratory
Chapter 22: Endocrine system

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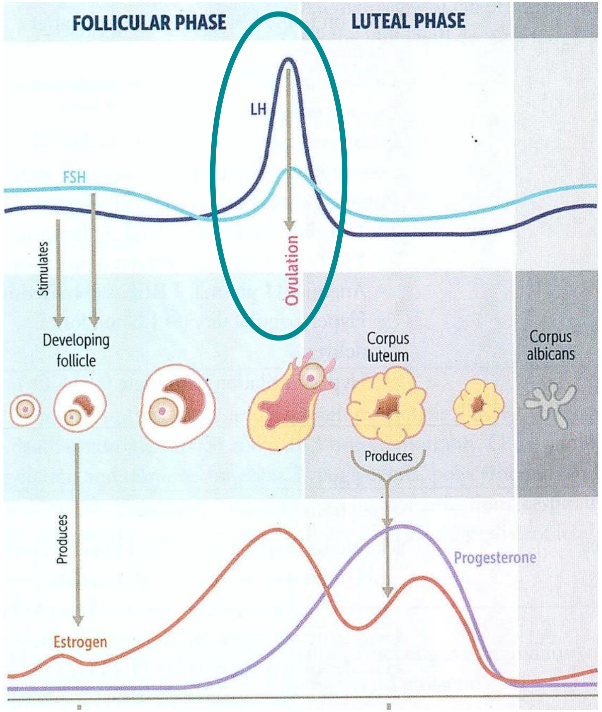
Menstrual cycle: Luteinizing hormone (LH)



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Early follicular phase:

- LH secretion pulses ~ every 1 hour (constant amplitude)

Prior to ovulation:

- LH pulse frequency increases (conc increase).
- Estrogen secretion from gravid follicle peak to 150 -200 pg/ml for ~36 -50 hrs
- **Estrogen switches to a brief positive-feedback effect on the pituitary to trigger the preovulatory surge of LH and FSH.**

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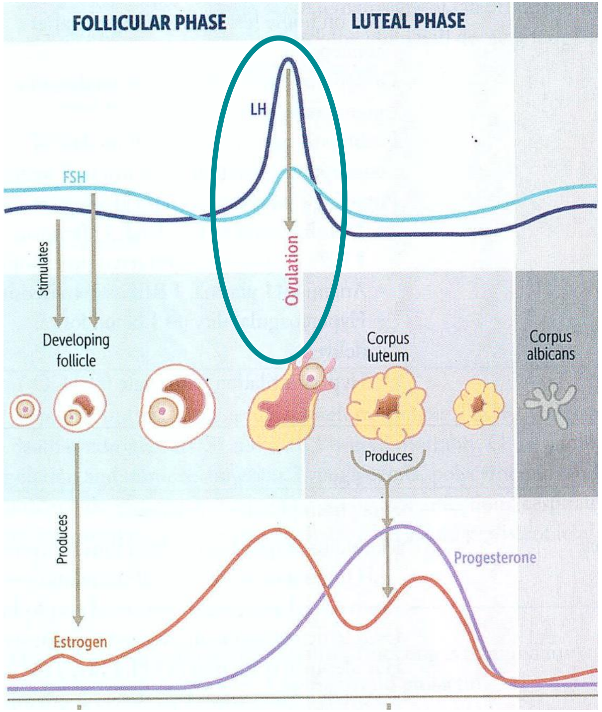
Laboratory Medicine: The Diagnosis of Disease in the Clinical Laboratory Chapter 22: Endocrine system

Menstrual cycle: Luteinizing hormone (LH) ctd



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Prior to ovulation:

- Increased estrogen stimulates LH in a surge ~ 10 to 12 hrs before ovulation.
- **LH surge** → Luteinization of granulosa cells & stimulates synthesis of progesterone responsible for the midcycle FSH surge
- **LH surge** also → rupture & ovulation (enzymes and hyperemia of follicle)

[Laboratory Medicine: The Diagnosis of Disease in the Clinical Laboratory](#) Chapter 22: Endocrine system

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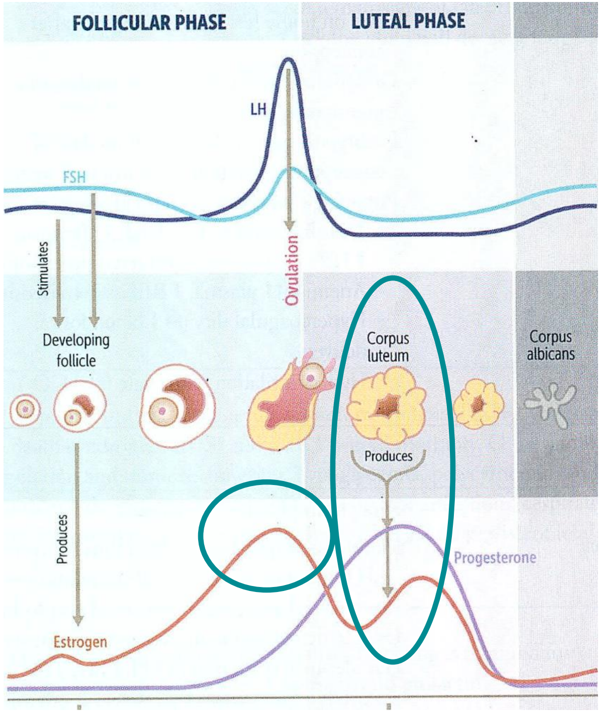
Menstrual cycle: Estrogen and progesterone



Modified from First Aid 2016 USLME Step 1. McGraw Hill

ESTROGEN PEAKS TWICE IN CYCLE

Follicular phase: secreted by follicle. Levels proportional follicle size
(# of granulosa cells)

Luteal phase: Corpus luteum secretes estradiol & also progesterone to facilitate implantation.

If no fertilization:

- Corpus luteum breaks down
- Estradiol & progesterone synth decline
- Menstrual cycle begins again.

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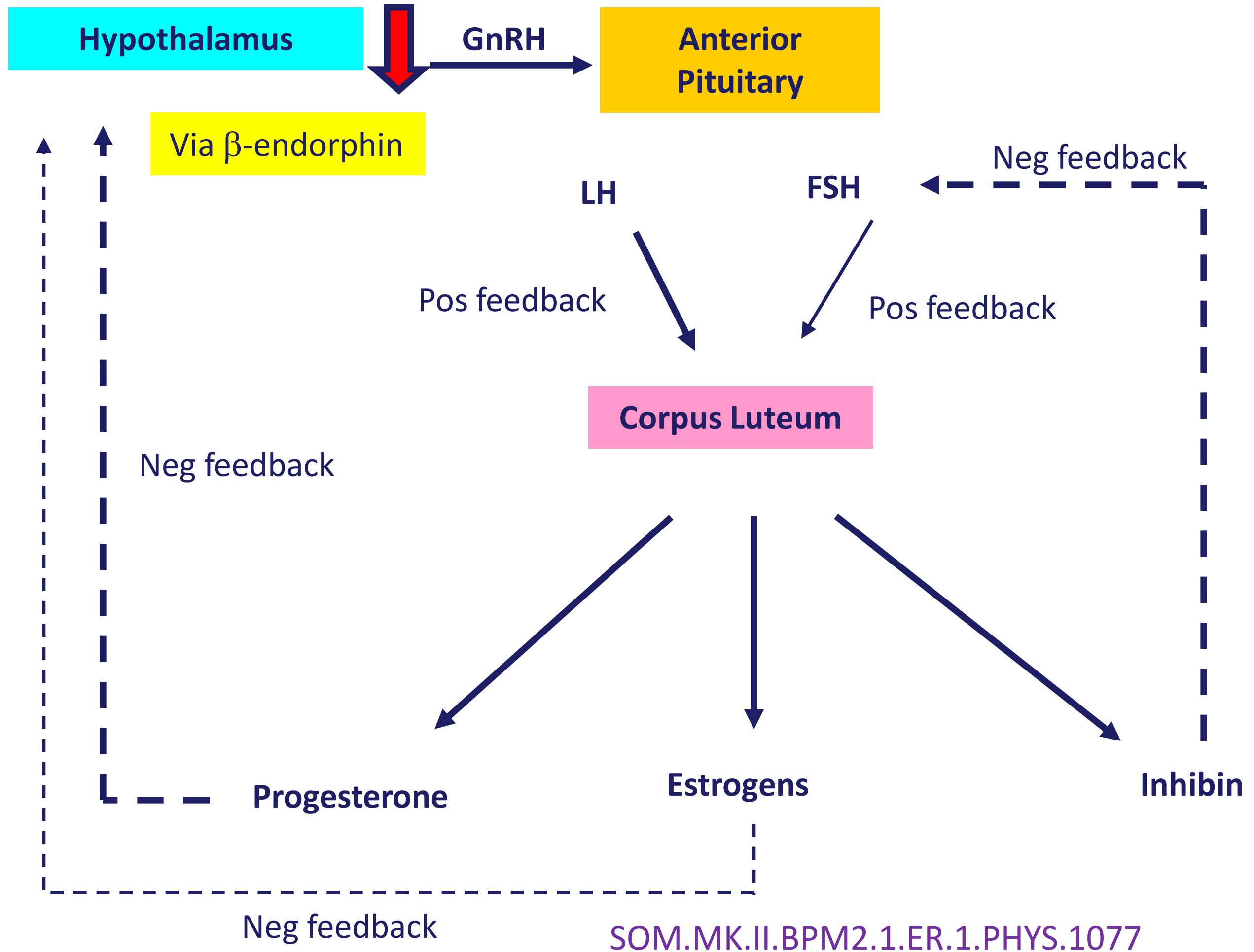
Menstrual cycle: Corpus Luteum



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Ovarian Hormones

Estrogens

Estrogens

- For the majority of the cycle, the reproductive system functions in a classic endocrine negative feedback mode.

- Estrogens are a group of steroidal hormones secreted in large quantities by the ovaries
  - β estradiol is the major hormone secreted by the ovaries in a non-pregnant female
  - 12 times more potent than estrone
  - 80 times more potent than estriol

- Is carried in blood bound to albumin & sex hormone binding globulin(SHBG) proteins (loosely bound)

- Is metabolized by the liver

Function of estrogens

Functions on ovary & endometrium: Ovarian & menstrual function, prepare endometrium at puberty

Functions on reproductive organs- Initiation of breast development at puberty & ductal proliferation, makes vaginal epithelium resistant to trauma, increase size of reproductive organs, increase number & activity of cilia in oviducts

Functions on other organ/system- Causes H_2O retention, protein synthesis, sex binding proteins, subcutaneous fat deposition (e.g. hips), epiphyseal fusion, builds bone, decreases cholesterol levels

Progesterone

Progesterone

- Progesterone secreted in large amounts by
  - corpus luteum during luteal phase
  - placenta
- Transported similarly to estrogens
- Liver very quickly breaks down to inactive forms

Functions of Progesterone

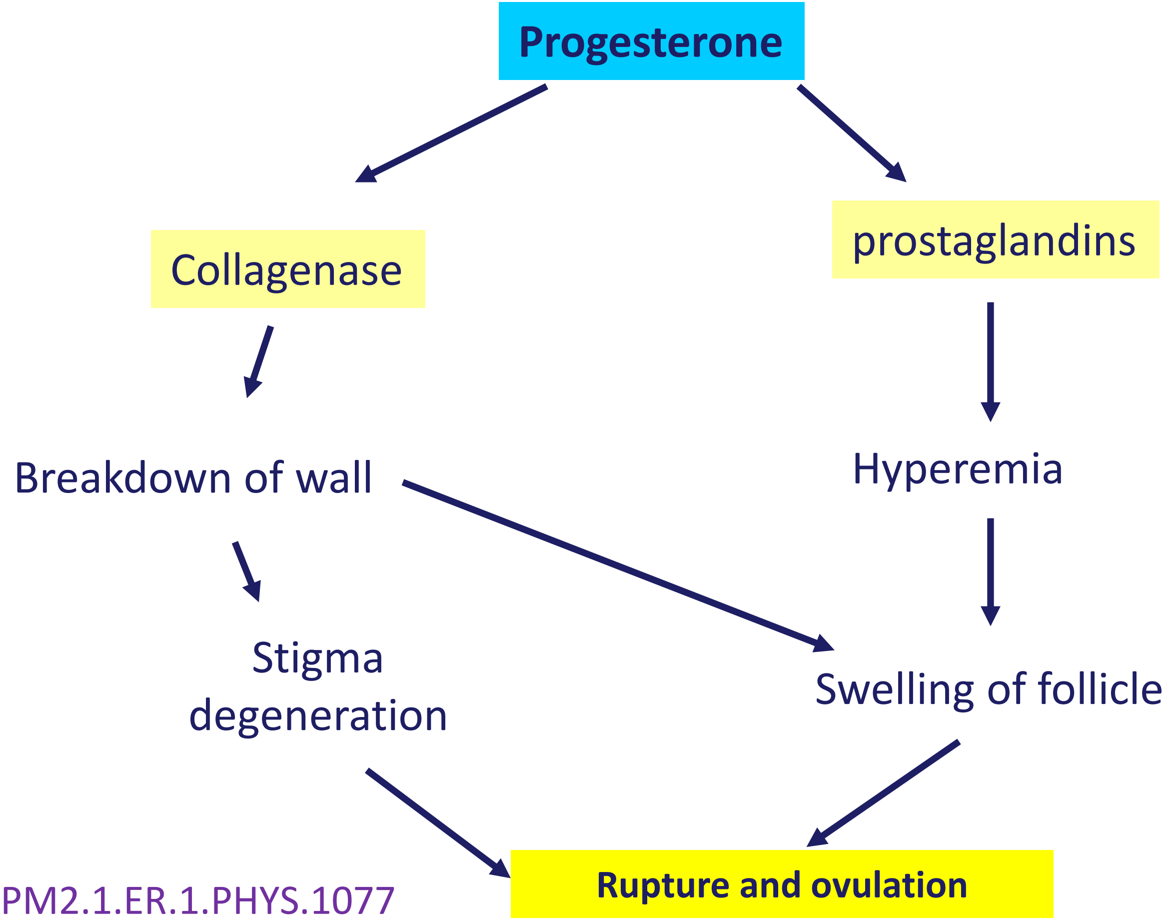
- Stimulates endometrial glandular secretions and prepares endometrium for implantation of the fertilized egg
- Aids ovulation
- Promotes development of lobules & alveoli in the breast & glandular proliferation
- In pregnancy - inhibits uterine excitability

Role of progesterone- Just prior to ovulation

Progesterone

- Helps LH surge
- Role in FSH surge
- Inhibits estrogen
- Ovulation processes

Role of progesterone- ovulation



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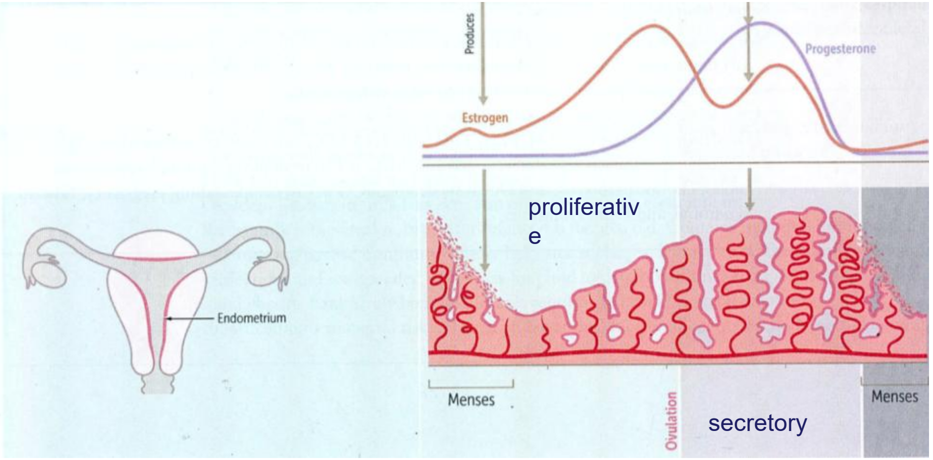
Uterine Cycle

- The uterine endometrium, where a fertilized egg will implant, also undergoes cyclical changes.

- It has 3 distinctive phases (even though there 4 phases – the ischemic phase)
  - menstrual
  - proliferative
  - secretory

- These phases are dictated by ovarian steroids (estradiol and progesterone)

Uterine Cycle



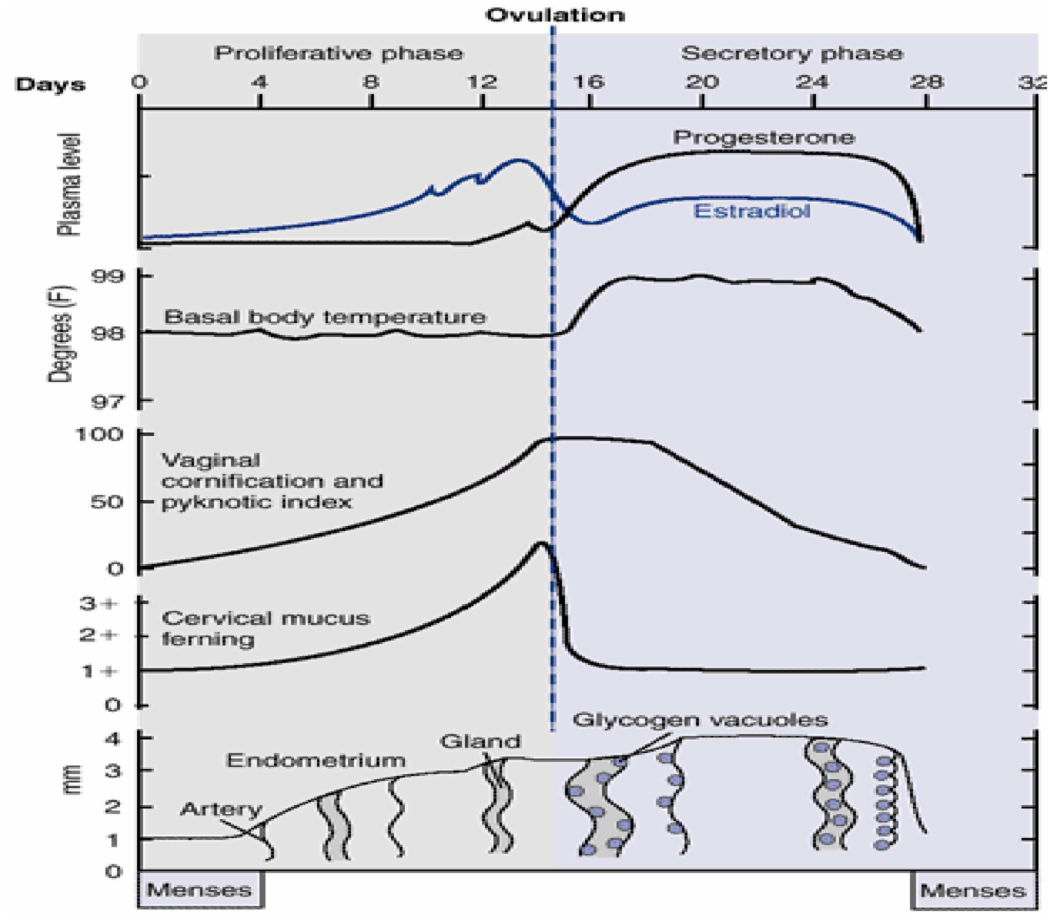
First Aid 2016 USLME Step 1. McGraw Hill

Menses phases (2-7 days)
- Shedding of endometrium (preceded by short ischemic phase)
- Due to abrupt withdrawal progesterone
- Within follicular phase of menstrual cycle; progesterone, FSH, & LH are low

Proliferative phase:
- Changes in vagina and thickening of uterus (epithelium, mucus, etc)
- Due to increased amounts of estrogen secreted from follicle.

Secretory phase:
- Preparation for implantation
- Increase in glycogen secreting glands.

Uterine cycle- hormone & other changes



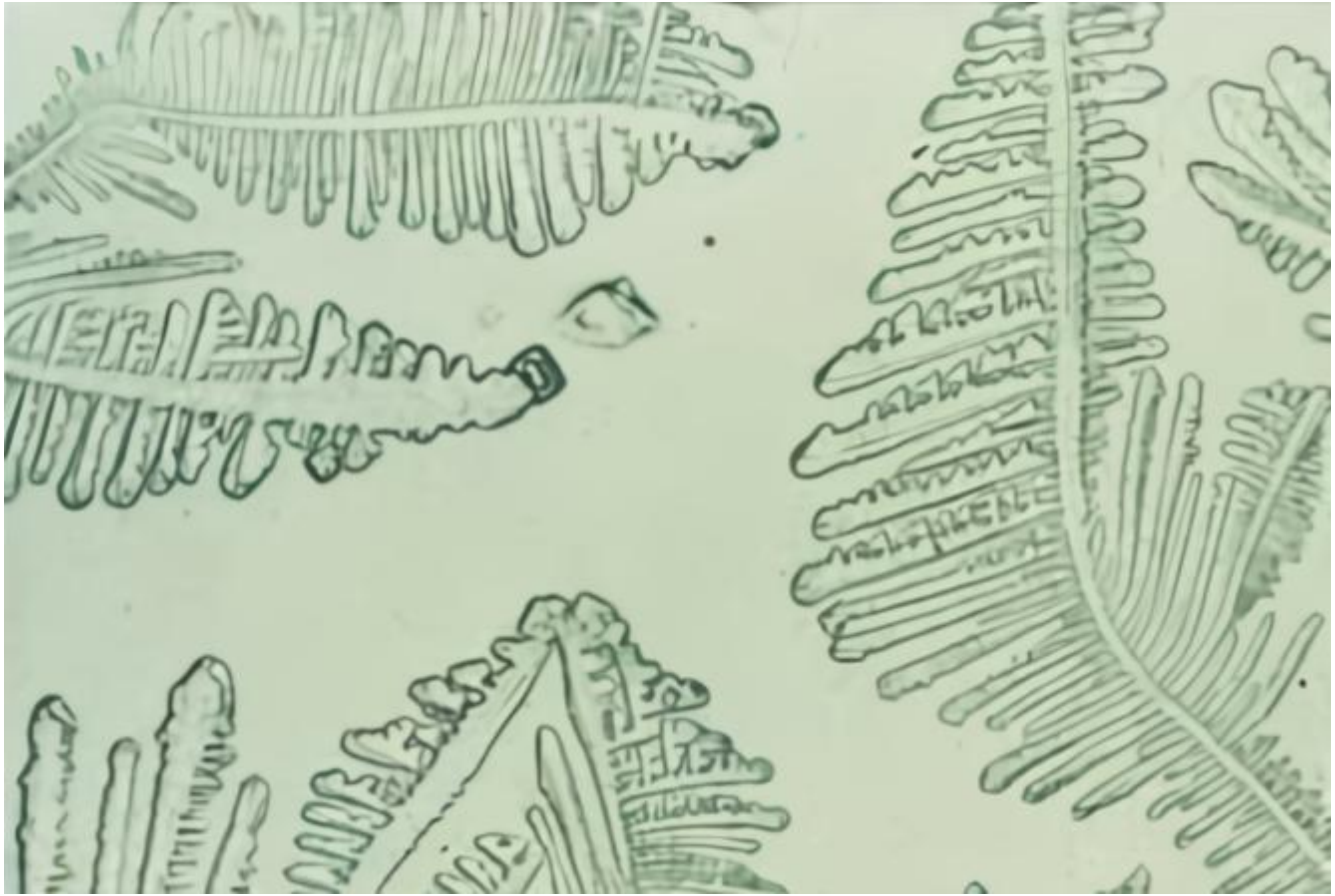
Ovarian hormones important

Temp increase

Keratinization.

At ovulation- mucus becomes thinner to allow passage of sperm

Cervical mucus ferning test for Ovulation



Age- related changes:

- puberty
- reproductive maturity
- reproductive senescence (menopause)

Age- related changes:

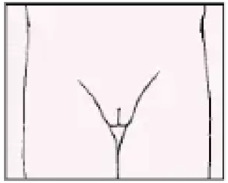
Tanner classification of female adolescent development.

Age	**Stage**	**Breast Development**	**Pubic Hair Development**
Pre-pubertal	I	Papillae elevated (preadolescent), no breast buds	None
~ 8 - 11.5 years	II	Breast buds and papillae slightly elevated	Sparse, long, slightly pigmented
~ 11.5–13 years	III	Breasts and areolae confluent, elevated	Darker, coarser, curly
~ 13–15 years	IV	Areolae and papillae project above breast	Adult-type pubis only
Usually, > 15 years	V	Papillae projected, mature	Lateral distribution

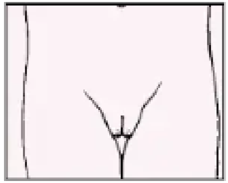
CURRENT Diagnosis & Treatment: Obstetrics & Gynecology, 12e

Tanner Staging

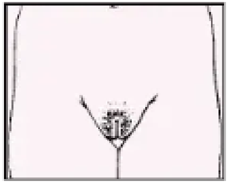
Staging Pubic Hair Development



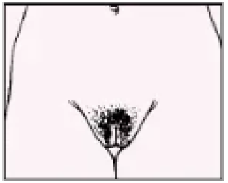
Pubic Hair Stage 1
Prepubertal. The vellus over the pubis is no further developed than that over the abdominal wall, ie, no pubic hair.



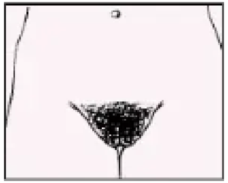
Pubic Hair Stage 2
Sparse growth of slightly pigmented, longer but still downy hair, straight or only slightly curled, appearing chiefly along the labia.



Pubic Hair Stage 3
The hair is considerably darker, coarser, and more curled. The hair spreads sparsely over the mons.

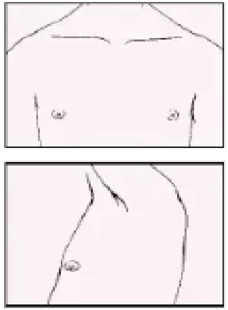


Pubic Hair Stage 4
The hair now resembles adult type. The area covered is still smaller than that in the adult, but the hair is beginning to spread across the mons. There is no hair spread to the medial thighs.

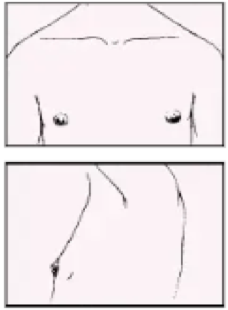


Pubic Hair Stage 5
The hair is adult in type and quantity; darker, coarse, and curled; and distributed in the classic female triangle. Some individuals may have hair spread to the medial thighs.

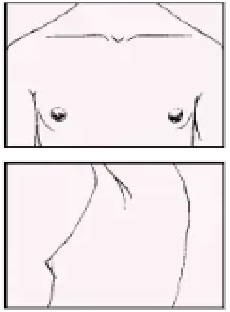
Staging Breast Development



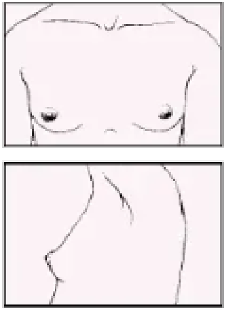
Breast Stage 1
There is no development. Only the papilla is elevated.



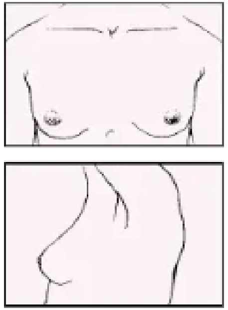
Breast Stage 2
The "breast bud" stage. The areola widens, darkens slightly, and elevates from the rest of the breast as a small mound. A bud of breast tissue is palpable below the nipple.



Breast Stage 3
The breast and areola further enlarge and present a rounded contour. There is no separation of contour between the nipple and areola and the rest of the breast. The breast tissue creates a small cone.

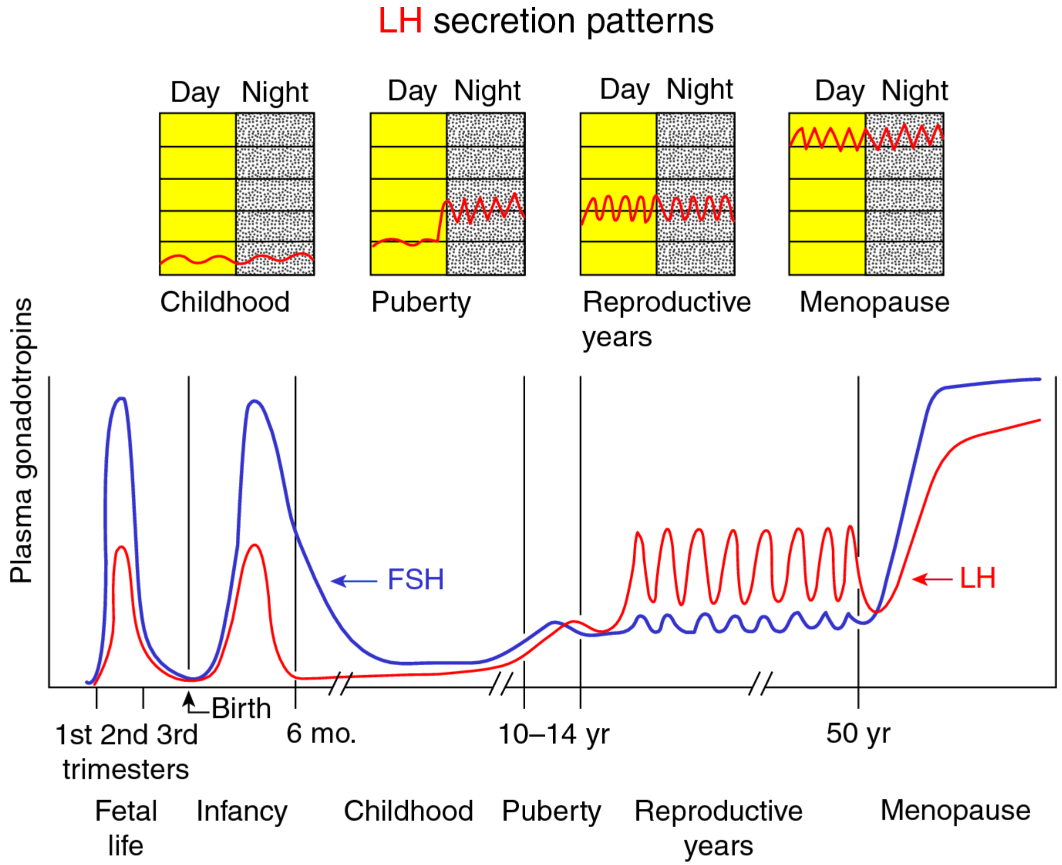


Breast Stage 4
The breast continues to expand. The papilla and areola project to form a secondary mound above the rest of the breast tissue.



Breast Stage 5
The mature adult stage. The secondary mound made by the areola and nipple, present in Stage 4, disappears. Only the papilla projects. The diameter of the breast tissue (as opposed to the height) has extended to cover most of the area between the sternum and lateral chest wall.

Puberty and Menopause



Menopause
↑in FSH & LH
FSH rise> LH

SOM.MK.II.BPM2.1.ER.1.PHYS.1081

Source: Barbara L. Hoffman, John O. Schorge, Karen D. Bradshaw, Lisa M. Halvorson, Joseph I. Schaffer, Marlene M. Corton: *Williams Gynecology*, 3rd Edition: www.accessmedicine.com

Williams Gynecology, 3e > Reproductive Endocrinology
https://accessmedicine.mhmedical.com/ViewLarge.aspx?figid=118169711&gbosContainerID=0&gbosid=0&groupID=0

Puberty and Menopause

Puberty (11- 13 years, start is called menarche)

- GnRH activity increases
- Eventually leads to ovarian activity
- Increased estrogen levels stimulate reproductive organ development.

Menopause (45 – 52 years)

- Cessation of the ovarian cycle
- Decreased estrogen secretion.
- Ovariectomy or ovary lack of function-
  - Overproduction of FSH and LH

Menopause - symptoms

- Around start of menopause the levels of estrogens decline leading to:

- Vasomotor symptoms such as
  - hot flashes,
  - irritability,
  - anxiety & depression

- Atrophy of estrogen dependent tissues
  - atrophy of vaginal epithelium
  - osteoporosis

Birth control-Hormonal suppression of fertility

- Prevention of ovulation: Estrogen or progesterone in sufficient quantities during the 1st half of the ovarian cycle
- Both hormones will suppress the LH surge required for ovulation

- Most common estrogens are
  - ethinyl estradiol, estradiol & mestranol
- Most common progestins are
  - norethindrone & norgestrel

Clinical correlations

Hypogonadism: ovaries absent/non-functional

Before puberty: female eunuchism

- No female secondary sexual characteristics
- No development of sexual organs
- Epiphyseal closure of their long bones is delayed & hence they are taller

Hypogonadism: Post puberty

- Regression of reproductive organs (esp. uterus)
- Vaginal epithelium easily damaged
- Irregularities in menstrual cycle / amenorrhea
- Prolonged ovarian cycle